

```
In [35]: %pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from seaborn) (2.5.2)
Requirement already satisfied: pandas>=1.2 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from seaborn) (3.0.5)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from seaborn) (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.64.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.5.1)
Requirement already satisfied: packaging>=20.0 in C:\Users\Sahithya\AppData\Roaming\Python\Python314\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.3)
Requirement already satisfied: pillow>=9 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.3.0)
Requirement already satisfied: pyparsing>=3 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: tzdata in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pandas>=1.2->seaborn) (2026.3)
Requirement already satisfied: six>=1.5 in c:\Users\Sahithya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
[notice] A new release of pip is available: 26.1.2 -> 26.2.1
```

```
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
In [36]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [37]: df = sns.load_dataset("titanic")
```

```
In [38]: df.head()
```

Out[38]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T



```
In [39]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   survived        891 non-null    int64
1   pclass          891 non-null    int64
2   sex             891 non-null    str
3   age             714 non-null    float64
4   sibsp           891 non-null    int64
5   parch           891 non-null    int64
6   fare            891 non-null    float64
7   embarked        889 non-null    str
8   class           891 non-null    category
9   who             891 non-null    str
10  adult_male      891 non-null    bool
11  deck            203 non-null    category
12  embark_town     889 non-null    str
13  alive           891 non-null    str
14  alone           891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
memory usage: 80.7 KB
```

```
In [40]: df.describe()
```

Out[40]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [41]: df["survived"].value_counts()
```

```
Out[41]: survived
0      549
1      342
Name: count, dtype: int64
```

```
In [42]: df.isnull().sum()
```

```
Out[42]: survived      0
pclass      0
sex      0
age      177
sibsp      0
parch      0
fare      0
embarked      2
class      0
who      0
adult_male      0
deck      688
embark_town      2
alive      0
alone      0
dtype: int64
```

```
In [43]: df.shape
```

```
Out[43]: (891, 15)
```

```
In [44]: df.isnull().sum()
```

```
Out[44]: survived      0
pclass      0
sex      0
age      177
sibsp      0
parch      0
fare      0
embarked      2
class      0
who      0
adult_male      0
deck      688
embark_town      2
alive      0
alone      0
dtype: int64
```

```
In [45]: df["age"] = df["age"].fillna(df["age"].median())

df["embarked"] = df["embarked"].fillna(df["embarked"].mode()[0])

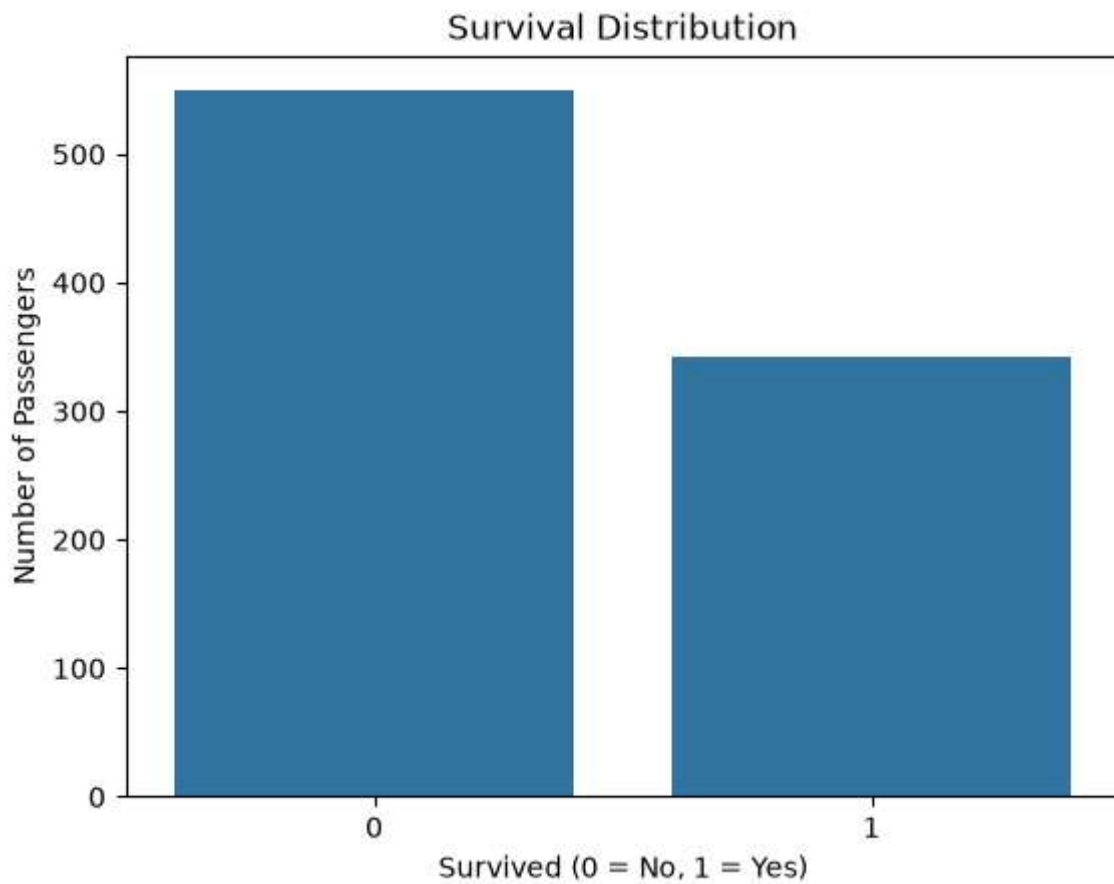
df["embark_town"] = df["embark_town"].fillna(df["embark_town"].mode()[0])
```

```
df = df.drop(columns=["deck"])
```

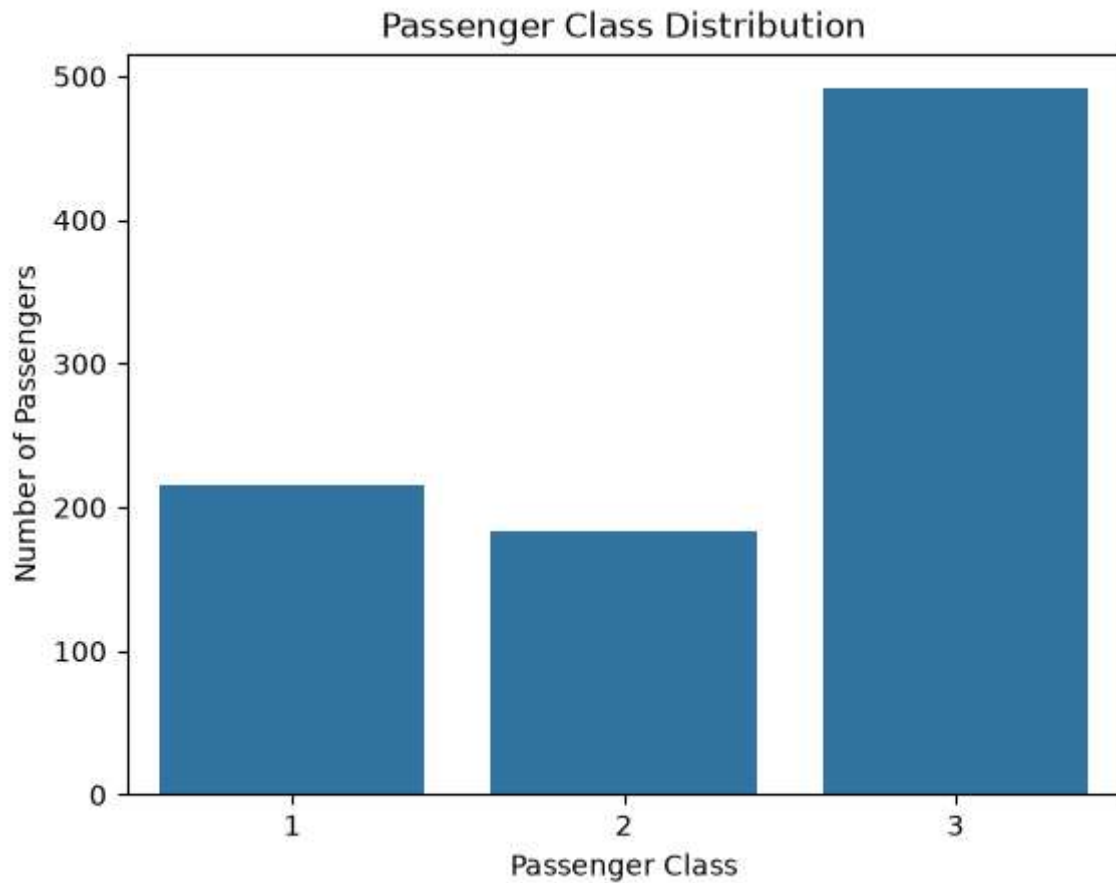
```
In [46]: df.isnull().sum()
```

```
Out[46]: survived      0  
pclass      0  
sex         0  
age         0  
sibsp       0  
parch       0  
fare        0  
embarked    0  
class       0  
who         0  
adult_male  0  
embark_town  0  
alive       0  
alone       0  
dtype: int64
```

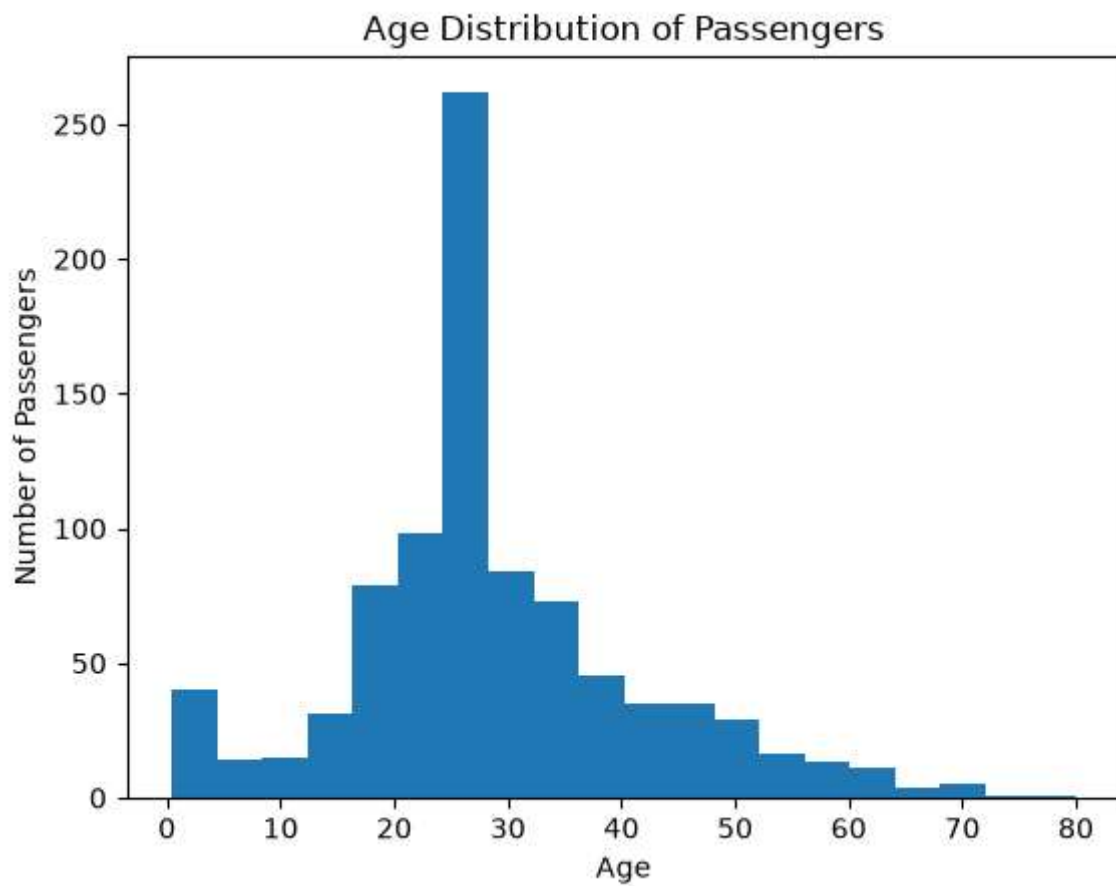
```
In [47]: sns.countplot(x="survived", data=df)  
plt.title("Survival Distribution")  
plt.xlabel("Survived (0 = No, 1 = Yes)")  
plt.ylabel("Number of Passengers")  
plt.show()
```



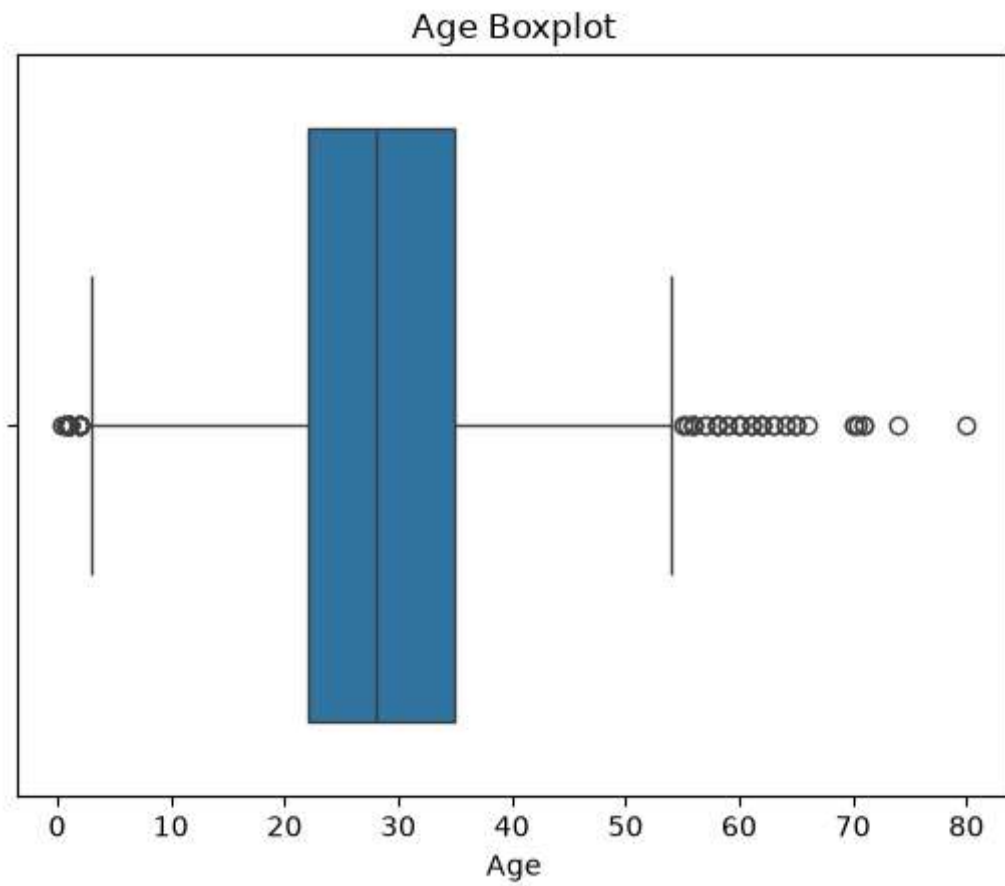
```
In [48]: sns.countplot(x="pclass", data=df)
plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()
```



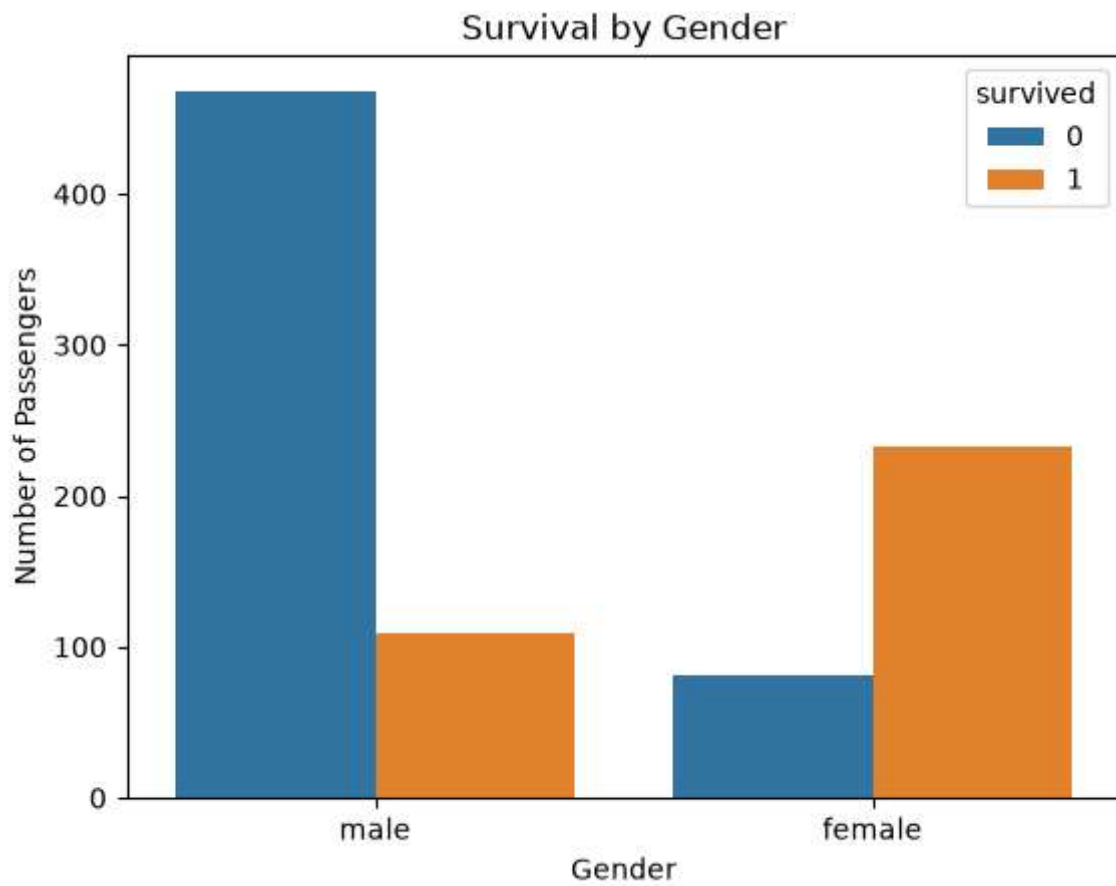
```
In [49]: plt.hist(df["age"], bins=20)
plt.title("Age Distribution of Passengers")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()
```



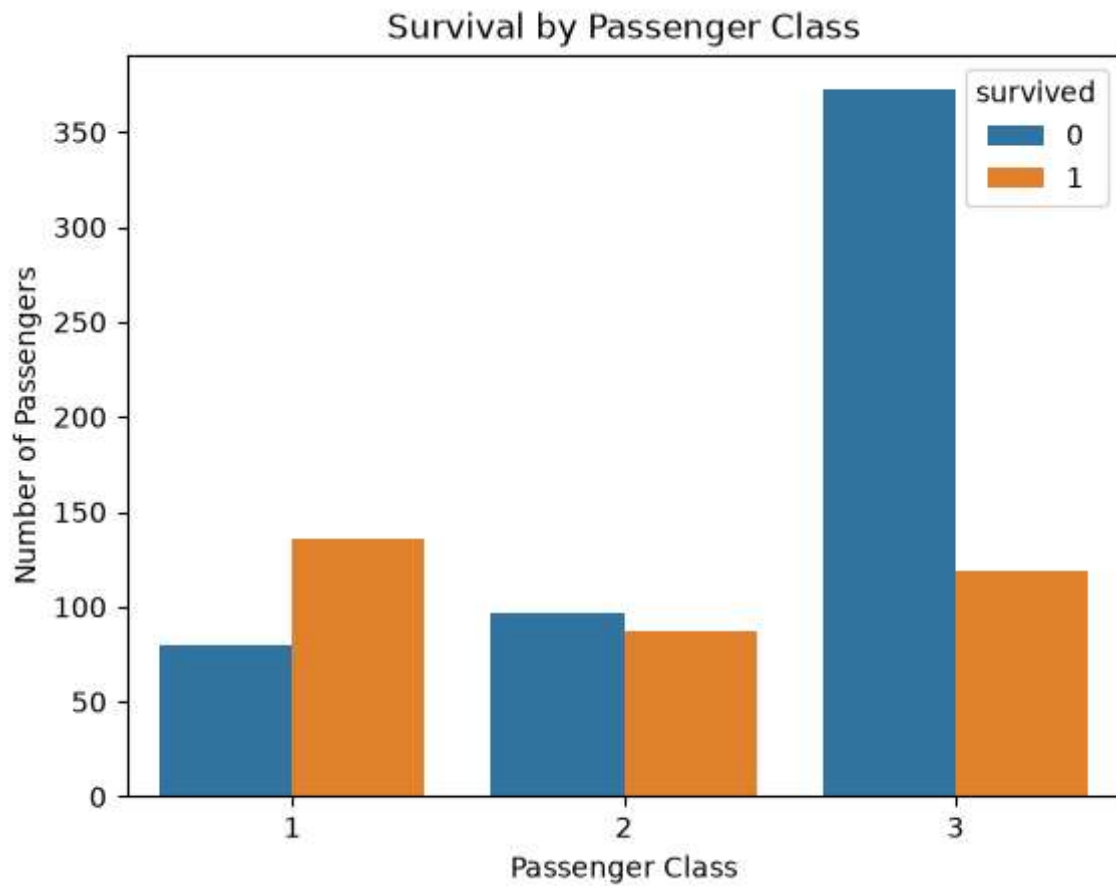
```
In [50]: sns.boxplot(x=df["age"])  
plt.title("Age Boxplot")  
plt.xlabel("Age")  
plt.show()
```



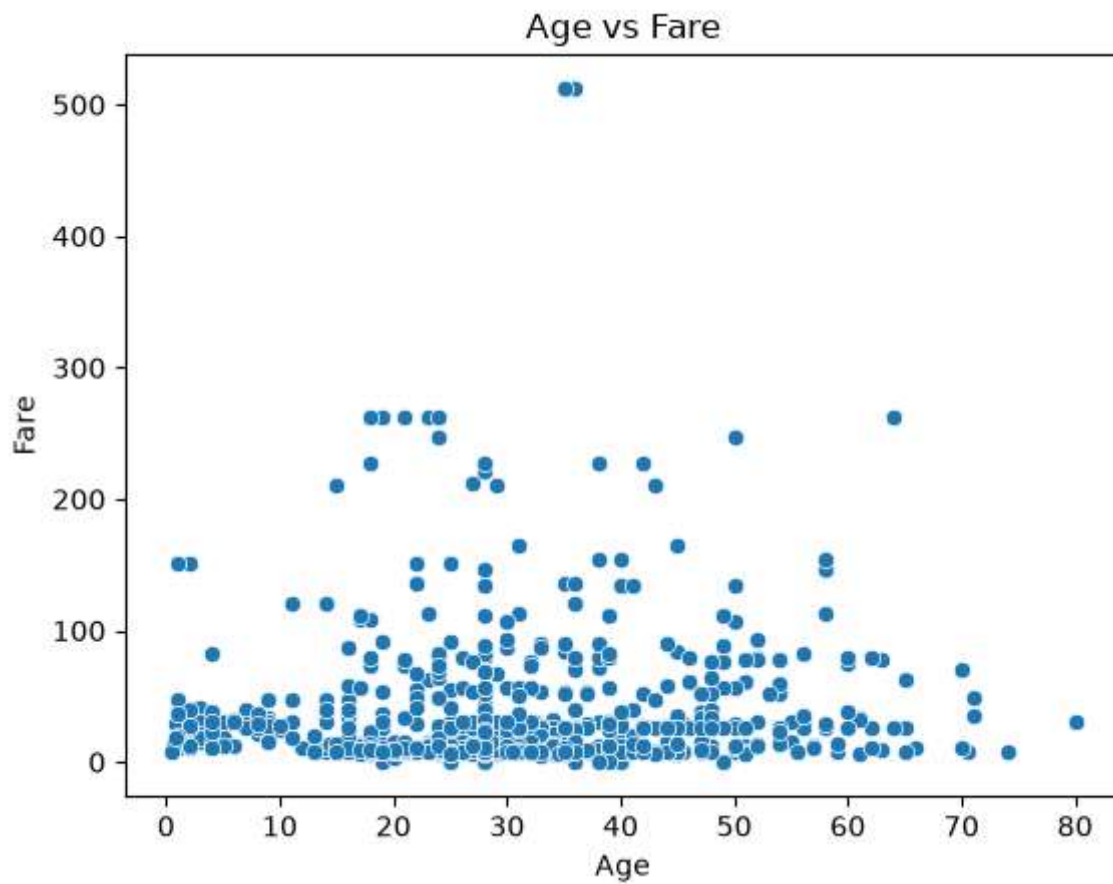
```
In [51]: sns.countplot(x="sex", hue="survived", data=df)
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```



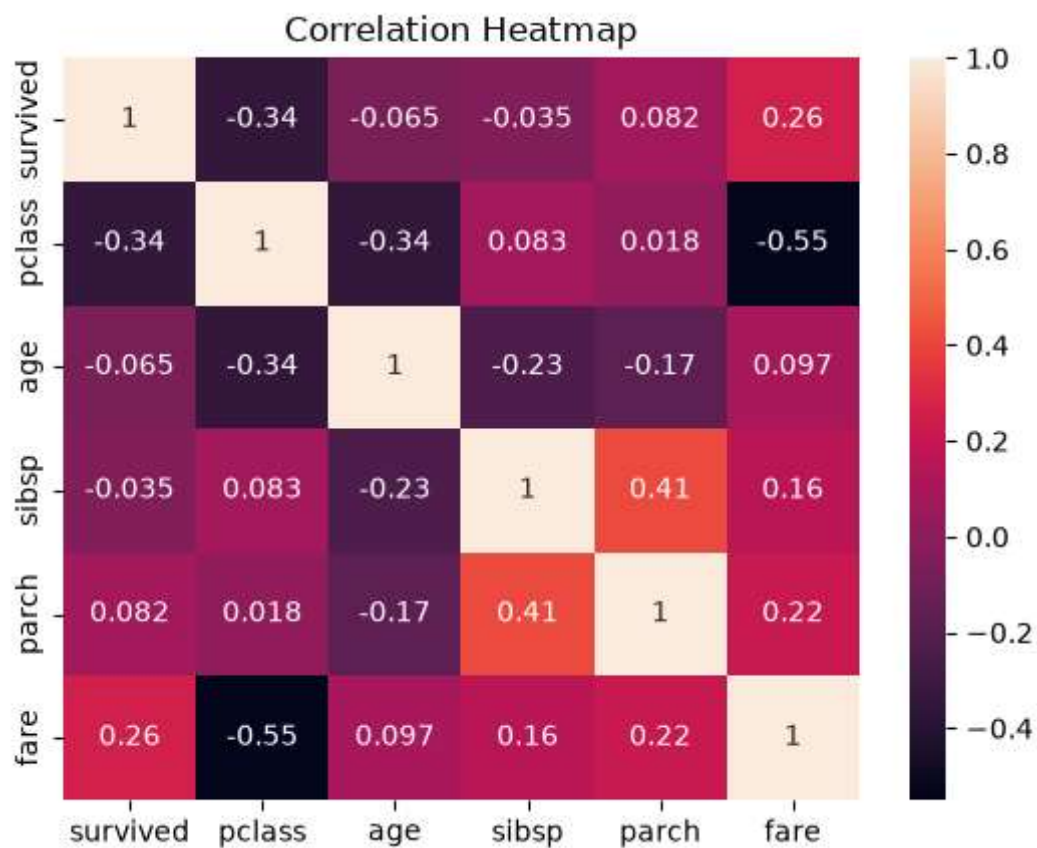
```
In [52]: sns.countplot(x="pclass", hue="survived", data=df)
plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()
```



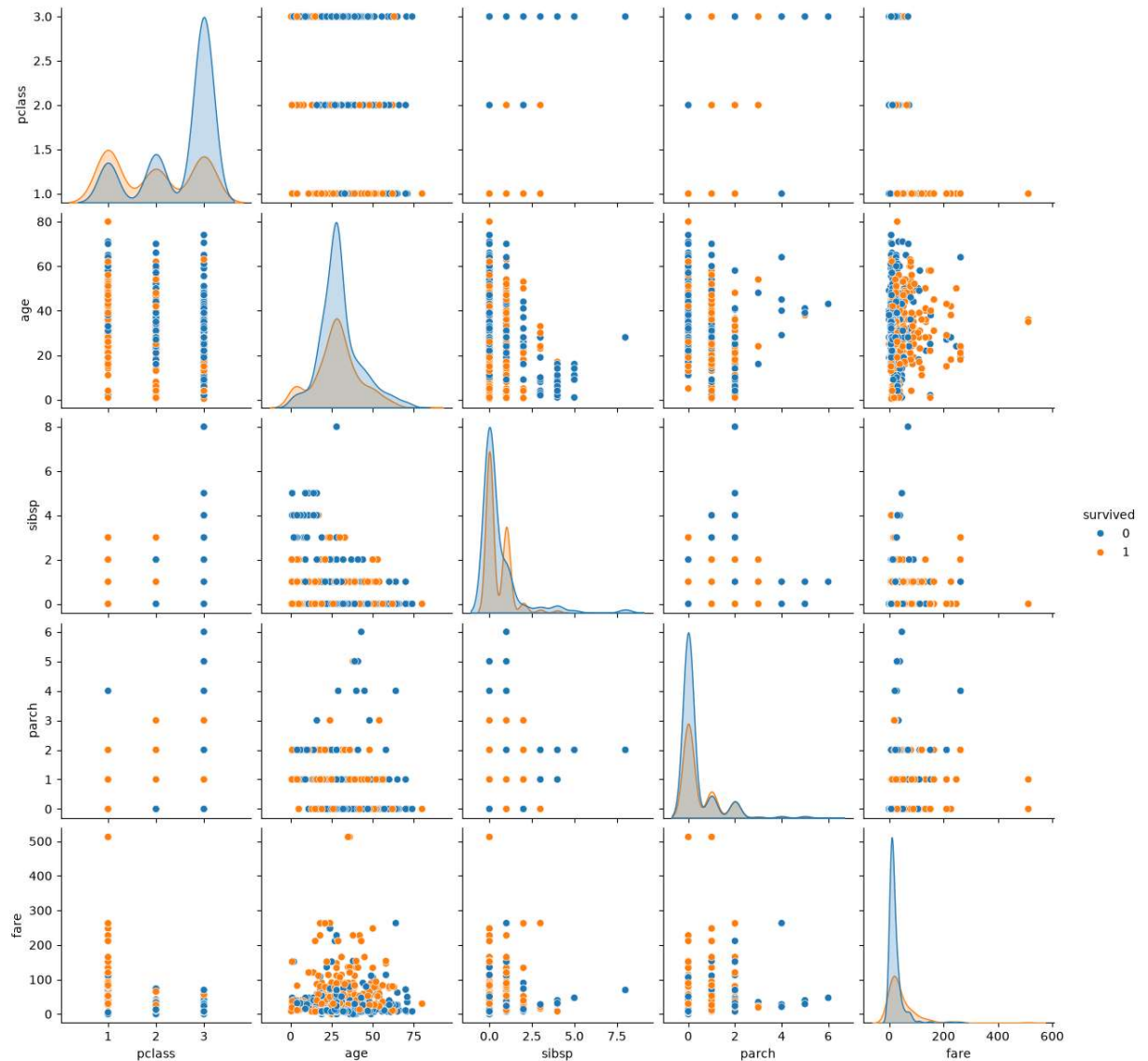
```
In [53]: sns.scatterplot(x="age", y="fare", data=df)
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```



```
In [54]: correlation = df.select_dtypes(include="number").corr()  
  
sns.heatmap(correlation, annot=True)  
plt.title("Correlation Heatmap")  
plt.show()
```



```
In [55]: sns.pairplot(  
    df[["survived", "pclass", "age", "sibsp", "parch", "fare"]],  
    hue="survived"  
)  
plt.show()
```



Observations

1. Survival Distribution

The number of passengers who did not survive was higher than the number of passengers who survived.

2. Passenger Class Distribution

Third-class passengers formed the largest group of passengers, followed by first-class and second-class passengers.

3. Age Distribution

Most passengers were adults, while fewer passengers were children and older adults.

4. Age Boxplot

The boxplot shows the distribution and spread of passenger ages and highlights some higher-age values as potential outliers.

5. Survival by Gender

Female passengers had a higher survival rate compared with male passengers.

6. Survival by Passenger Class

First-class passengers generally had better survival outcomes than second- and third-class passengers.

7. Age vs Fare

The scatterplot shows that passenger fare varies considerably across different ages, with some passengers paying much higher fares.

8. Correlation Heatmap

The heatmap shows relationships between the numerical variables. Passenger class and fare show a noticeable relationship with survival.

9. Pairplot

The pairplot provides an overall view of relationships among the numerical variables and shows differences between survived and non-survived passengers.

Summary of Findings

The Exploratory Data Analysis of the Titanic dataset revealed several important patterns.

Survival was not evenly distributed among passengers. Gender and passenger class showed noticeable differences in survival outcomes. Female passengers generally had better survival outcomes than male passengers, while first-class passengers generally had better survival outcomes than passengers in lower classes.

The analysis also showed variation in passenger ages and fares. The correlation heatmap and pairplot helped identify relationships among the numerical variables.

Overall, EDA helped identify important patterns, relationships, and possible outliers in the Titanic dataset.